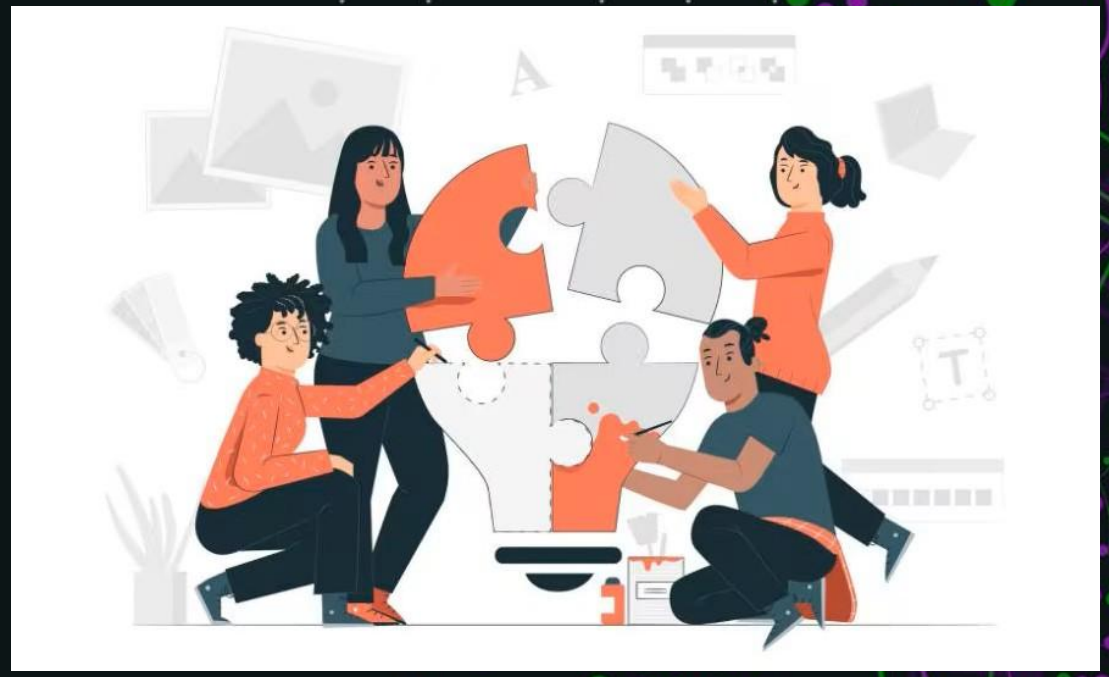
From Testing to Code

Writing unit code for functional requirements in Python

In the previous video...



In this video...

Recap

- Arithmetic Calculations:
  - Perform basic arithmetic functions:
    - Perform addition
    - Perform subtraction
    - Perform multiplication
    - Perform division

- Input Handling:
  - Accept input
  - Input must be 1 or 2 digits
  - Input must contain valid hexadecimal numbers
  - Input must include a valid operator (+, -, /, *)

- Output Handling:
  - Return and display output
  - Output must be less than 5 digits (<=4)
  - Output must not be negative
  - Output must not contain decimal places

- Error Handling:
  - Invalid hexadecimal input must return an error
  - Missing or invalid operator must return an error
  - Input exceeding 2 digits must return an error
  - Division by zero must return an error
  - Output that is greater than 4 digits will return an error
  - Output that is negative will return an error
  - Output with decimal places will return an error

Basic arithmetic

- What do we mean by that?
- Addition:
  - value_one + value_two = result
- Subtraction:
  - value_one - value_two = result
- Multiplication:
  - value_one * value_two = result
- Division:
  - value_one / value_two = result

Input Handling

- What do we mean by that?
- value_one and value_two:
  - must be 1 or 2 digits
  - must contain valid hexadecimal numbers
  - must include a valid operator (+, -, /, *)

Output Handling

- What do we mean by that?
- result:
  - must be less than 5 digits (<=4)
  - must not be negative
  - must not contain decimal places

Error Handling

- What do we mean by that?
- Input handling & Output handling:
  - Invalid hexadecimal input must return an error
  - Missing or invalid operator must return an error
  - Input exceeding 2 digits must return an error
  - Division by zero must return an error
  - Output that is greater than 4 digits will return an error
  - Output that is negative will return an error
  - Output with decimal places will return an error

In Python

```python
def function_name(parameters):
    """
    Brief description of what the function does.

    Args:
        parameters (type): Description of the parameters.

    Returns:
        type: Description of the return value.
    """

    # --- Function Logic ---


    # --- Return Result ---
    return result
```

Combining

- Arithmetic (addition), input, output and error handling:

```
def perform_addition(value_one, value_two):
    if not valid_hexadecimal(value_one) or not valid_hexadecimal(value_two): # Input handling
        raise ValueError("Input for addition must be a valid hexadecimal bewteen 1 and 2 digits") # Error
Handling

    decimal_one = int(value_one, 16)
    decimal_two = int(value_two, 16)

    result_decimal = decimal_one + decimal_two # Basic Arithmetic (addition)

    final_result = hex(result_decimal)[2:].upper() # Output Handling

    return final_result # Output Handling
```

Thank you for watching